CN Orange Problem

Title: SDN based firewall

Name: Naija Hodigere **SRN:** PES2UG24CS296 **Sec:** E

Abstract:

This project implements a controller-based firewall using Software Defined Networking (SDN) principles with Mininet and a Ryu controller. The objective is to demonstrate dynamic traffic control using OpenFlow rules by allowing or blocking communication between hosts based on predefined policies. The controller monitors incoming packets and applies match–action logic to enforce firewall rules.

A learning switch mechanism is integrated to enable normal forwarding behavior, while specific IP-based rules are used to block traffic between selected hosts. When a packet matches the defined blocked source–destination pair, the controller installs a high-priority drop rule in the switch, preventing further communication between those hosts. All other traffic is handled using standard forwarding rules.

Github link: sdn based firewall

Readme link: README

Screenshots and graphs:

1. Controller startup (terminal 1)

```
(ryu-env) naija@VelvetThunder:~/ryu$ python ryu/cmd/manager.py "/mnt/c/Users/naija/pes/sem 4/cn/firewall_controller.py"
/home/naija/ryu/ryu/lib/hub.py:30: EventletDeprecationWarning:
Eventlet is deprecated. It is currently being maintained in bugfix mode, and
we strongly recommend against using it for new projects.

If you are already using Eventlet, we recommend migrating to a different
framework.  For more detail see
https://eventlet.readthedocs.io/en/latest/asyncio/migration.html

  import eventlet
2 RLock(s) were not greened, to fix this error make sure you run eventlet.monkey_patch() before importing any other modu
les.
loading app /mnt/c/Users/naija/pes/sem 4/cn/firewall_controller.py
loading app ryu.controller.ofp_handler
instantiating app /mnt/c/Users/naija/pes/sem 4/cn/firewall_controller.py of FirewallController
instantiating app ryu.controller.ofp_handler of OFPHandler
```

2. Mininet topology (terminal 2)

```
naija@VelvetThunder:~$ sudo mn --topo single,3 --controller remote
*** Creating network
*** Adding controller
Unable to contact the remote controller at 127.0.0.1:6653
Unable to contact the remote controller at 127.0.0.1:6633
Setting remote controller to 127.0.0.1:6653
*** Adding hosts:
h1 h2 h3
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1) (h3, s1)
*** Configuring hosts
h1 h2 h3
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Starting CLI:
```

3. Blocked ping (t – 2)

Ping:

```
naija@VelvetThunder:~$ sudo mn --topo single,3 --controller remote
*** Creating network
*** Adding controller
Connecting to remote controller at 127.0.0.1:6653
*** Adding hosts:
h1 h2 h3
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1) (h3, s1)
*** Configuring hosts
h1 h2 h3
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Starting CLI:
mininet> h1 ping h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
^C
--- 10.0.0.2 ping statistics ---
96 packets transmitted, 0 received, 100% packet loss, time 97258ms
```

Blocked response: (t – 1)

```
loading app /mnt/c/Users/naija/pes/sem 4/cn/firewall_controller.py
loading app ryu.controller.ofp_handler
instantiating app /mnt/c/Users/naija/pes/sem 4/cn/firewall_controller.py of FirewallController
instantiating app ryu.controller.ofp_handler of OFPHandler
Packet in: c2:2e:f4:35:a4:1e -> 33:33:ff:35:a4:1e (port 3)
Packet in: c2:df:48:ca:e6:15 -> 33:33:ff:ca:e6:15 (port 1)
Packet in: fe:7e:61:8c:2a:bd -> 33:33:ff:8c:2a:bd (port 2)
Packet in: c2:2e:f4:35:a4:1e -> 33:33:00:00:00:16 (port 3)
Packet in: fe:7e:61:8c:2a:bd -> 33:33:00:00:00:16 (port 2)
Packet in: c2:2e:f4:35:a4:1e -> 33:33:00:00:00:02 (port 3)
Packet in: c2:df:48:ca:e6:15 -> 33:33:00:00:00:16 (port 1)
Packet in: c2:df:48:ca:e6:15 -> 33:33:00:00:00:02 (port 1)
Packet in: fe:7e:61:8c:2a:bd -> 33:33:00:00:00:02 (port 2)
Packet in: c2:df:48:ca:e6:15 -> ff:ff:ff:ff:ff:ff (port 1)
Packet in: fe:7e:61:8c:2a:bd -> c2:df:48:ca:e6:15 (port 2)
Packet in: c2:df:48:ca:e6:15 -> fe:7e:61:8c:2a:bd (port 1)
🚫 BLOCKED: 10.0.0.1 -> 10.0.0.2
XPacket in: c2:df:48:ca:e6:15 -> fe:7e:61:8c:2a:bd (port 1)
```

4. Allowed case (t – 2)

Ping + packets

```
mininet> h2 ping h3
PING 10.0.0.3 (10.0.0.3) 56(84) bytes of data.
64 bytes from 10.0.0.3: icmp_seq=1 ttl=64 time=15.8 ms
64 bytes from 10.0.0.3: icmp_seq=2 ttl=64 time=0.452 ms
64 bytes from 10.0.0.3: icmp_seq=3 ttl=64 time=0.172 ms
64 bytes from 10.0.0.3: icmp_seq=4 ttl=64 time=0.111 ms
64 bytes from 10.0.0.3: icmp_seq=5 ttl=64 time=0.103 ms
64 bytes from 10.0.0.3: icmp_seq=6 ttl=64 time=0.182 ms
64 bytes from 10.0.0.3: icmp_seq=7 ttl=64 time=0.091 ms
64 bytes from 10.0.0.3: icmp_seq=8 ttl=64 time=0.172 ms
64 bytes from 10.0.0.3: icmp_seq=9 ttl=64 time=0.169 ms
64 bytes from 10.0.0.3: icmp_seq=10 ttl=64 time=0.174 ms
64 bytes from 10.0.0.3: icmp_seq=11 ttl=64 time=0.171 ms
64 bytes from 10.0.0.3: icmp_seq=12 ttl=64 time=0.181 ms
64 bytes from 10.0.0.3: icmp_seq=13 ttl=64 time=0.174 ms
64 bytes from 10.0.0.3: icmp_seq=14 ttl=64 time=0.182 ms
64 bytes from 10.0.0.3: icmp_seq=15 ttl=64 time=0.172 ms
64 bytes from 10.0.0.3: icmp_seq=16 ttl=64 time=0.177 ms
64 bytes from 10.0.0.3: icmp_seq=17 ttl=64 time=0.142 ms
64 bytes from 10.0.0.3: icmp_seq=18 ttl=64 time=0.172 ms
64 bytes from 10.0.0.3: icmp_seq=19 ttl=64 time=0.140 ms
64 bytes from 10.0.0.3: icmp_seq=20 ttl=64 time=0.246 ms
64 bytes from 10.0.0.3: icmp_seq=21 ttl=64 time=0.104 ms
64 bytes from 10.0.0.3: icmp_seq=22 ttl=64 time=0.085 ms
64 bytes from 10.0.0.3: icmp_seq=23 ttl=64 time=0.231 ms
64 bytes from 10.0.0.3: icmp_seq=24 ttl=64 time=0.181 ms
64 bytes from 10.0.0.3: icmp_seq=25 ttl=64 time=0.169 ms
^C
--- 10.0.0.3 ping statistics ---
25 packets transmitted, 25 received, 0% packet loss, time 24547ms
```

Response: (t – 1)

```
Packet in: fe:7e:61:8c:2a:bd -> ff:ff:ff:ff:ff:ff (port 2)
Packet in: c2:2e:f4:35:a4:1e -> fe:7e:61:8c:2a:bd (port 3)
Packet in: fe:7e:61:8c:2a:bd -> c2:2e:f4:35:a4:1e (port 2)
```

5. Controller logs (t – 1)

```
(ryu-env) naija@VelvetThunder:~/ryu$ python ryu/cmd/manager.py "/mnt/c/Users/naija/pes/sem 4/cn/firewall_controller.py"
/home/naija/ryu/ryu/lib/hub.py:30: EventletDeprecationWarning:
Eventlet is deprecated. It is currently being maintained in bugfix mode, and
we strongly recommend against using it for new projects.

If you are already using Eventlet, we recommend migrating to a different
framework.  For more detail see
https://eventlet.readthedocs.io/en/latest/asyncio/migration.html

  import eventlet
2 RLock(s) were not greened, to fix this error make sure you run eventlet.monkey_patch() before importing any other modules.
loading app /mnt/c/Users/naija/pes/sem 4/cn/firewall_controller.py
loading app ryu.controller.ofp_handler
instantiating app /mnt/c/Users/naija/pes/sem 4/cn/firewall_controller.py of FirewallController
instantiating app ryu.controller.ofp_handler of OFPHandler
Packet in: 22:28:5c:f3:c3:59 -> 33:33:00:00:00:16 (port 2)
Packet in: 22:28:5c:f3:c3:59 -> 33:33:ff:f3:c3:59 (port 2)
Packet in: be:9b:3f:70:ed:aa -> 33:33:ff:70:ed:aa (port 3)
Packet in: ea:a1:ef:82:9d:b9 -> 33:33:00:00:00:16 (port 1)
Packet in: ea:a1:ef:82:9d:b9 -> 33:33:00:00:00:02 (port 1)
Packet in: 22:28:5c:f3:c3:59 -> 33:33:00:00:00:02 (port 2)
Packet in: be:9b:3f:70:ed:aa -> 33:33:00:00:00:16 (port 3)
Packet in: be:9b:3f:70:ed:aa -> 33:33:00:00:00:02 (port 3)
Packet in: ea:a1:ef:82:9d:b9 -> ff:ff:ff:ff:ff:ff (port 1)
Packet in: 22:28:5c:f3:c3:59 -> ea:a1:ef:82:9d:b9 (port 2)
Packet in: ea:a1:ef:82:9d:b9 -> 22:28:5c:f3:c3:59 (port 1)
🚫 BLOCKED: 10.0.0.1 -> 10.0.0.2
Packet in: ea:a1:ef:82:9d:b9 -> 22:28:5c:f3:c3:59 (port 1)
Packet in: 22:28:5c:f3:c3:59 -> ff:ff:ff:ff:ff:ff (port 2)
Packet in: be:9b:3f:70:ed:aa -> 22:28:5c:f3:c3:59 (port 3)
Packet in: 22:28:5c:f3:c3:59 -> be:9b:3f:70:ed:aa (port 2)
```

Logs of both allowed and blocked case

6. Flow table (t – 2)

```
mininet> sh ovs-ofctl dump-flows s1
 cookie=0x0, duration=330.917s, table=0, n_packets=95, n_bytes=9310, priority=100,ip,nw_src=10.0.0.1,nw_dst=10.0.0.2 actions=drop
 cookie=0x0, duration=336.479s, table=0, n_packets=0, n_bytes=0, priority=1,in_port="s1-eth3",dl_src=c2:2e:f4:35:a4:1e,dl_dst=33:33:ff:35:a4
:1e actions=FLOOD
 cookie=0x0, duration=336.447s, table=0, n_packets=0, n_bytes=0, priority=1,in_port="s1-eth1",dl_src=c2:df:48:ca:e6:15,dl_dst=33:33:ff:ca:e6
:15 actions=FLOOD
 cookie=0x0, duration=336.191s, table=0, n_packets=0, n_bytes=0, priority=1,in_port="s1-eth2",dl_src=fe:7e:61:8c:2a:bd,dl_dst=33:33:ff:8c:2a
:bd actions=FLOOD
 cookie=0x0, duration=336.063s, table=0, n_packets=2, n_bytes=180, priority=1,in_port="s1-eth3",dl_src=c2:2e:f4:35:a4:1e,dl_dst=33:33:00:00:
00:16 actions=FLOOD
 cookie=0x0, duration=336s, table=0, n_packets=2, n_bytes=180, priority=1,in_port="s1-eth2",dl_src=fe:7e:61:8c:2a:bd,dl_dst=33:33:00:00:00:1
6 actions=FLOOD
 cookie=0x0, duration=335.457s, table=0, n_packets=6, n_bytes=420, priority=1,in_port="s1-eth3",dl_src=c2:2e:f4:35:a4:1e,dl_dst=33:33:00:00:
00:02 actions=FLOOD
 cookie=0x0, duration=335.424s, table=0, n_packets=1, n_bytes=90, priority=1,in_port="s1-eth1",dl_src=c2:df:48:ca:e6:15,dl_dst=33:33:00:00:0
0:16 actions=FLOOD
 cookie=0x0, duration=335.424s, table=0, n_packets=6, n_bytes=420, priority=1,in_port="s1-eth1",dl_src=c2:df:48:ca:e6:15,dl_dst=33:33:00:00:
00:02 actions=FLOOD
 cookie=0x0, duration=335.168s, table=0, n_packets=6, n_bytes=420, priority=1,in_port="s1-eth2",dl_src=fe:7e:61:8c:2a:bd,dl_dst=33:33:00:00:
00:02 actions=FLOOD
 cookie=0x0, duration=330.922s, table=0, n_packets=0, n_bytes=0, priority=1,in_port="s1-eth1",dl_src=c2:df:48:ca:e6:15,dl_dst=ff:ff:ff:ff:ff
:ff actions=FLOOD
 cookie=0x0, duration=330.919s, table=0, n_packets=2, n_bytes=84, priority=1,in_port="s1-eth2",dl_src=fe:7e:61:8c:2a:bd,dl_dst=c2:df:48:ca:e
6:15 actions=output:"s1-eth1"
 cookie=0x0, duration=283.935s, table=0, n_packets=1, n_bytes=42, priority=1,in_port="s1-eth1",dl_src=c2:df:48:ca:e6:15,dl_dst=fe:7e:61:8c:2
a:bd actions=output:"s1-eth2"
 cookie=0x0, duration=189.599s, table=0, n_packets=0, n_bytes=0, priority=1,in_port="s1-eth2",dl_src=fe:7e:61:8c:2a:bd,dl_dst=ff:ff:ff:ff:ff
:ff actions=FLOOD
 cookie=0x0, duration=189.595s, table=0, n_packets=92, n_bytes=8624, priority=1,in_port="s1-eth3",dl_src=c2:2e:f4:35:a4:1e,dl_dst=fe:7e:61:8
c:2a:bd actions=output:"s1-eth2"
 cookie=0x0, duration=189.593s, table=0, n_packets=91, n_bytes=8526, priority=1,in_port="s1-eth2",dl_src=fe:7e:61:8c:2a:bd,dl_dst=c2:2e:f4:3
5:a4:1e actions=output:"s1-eth3"
 cookie=0x0, duration=3.655s, table=0, n_packets=16, n_bytes=1144, priority=0 actions=CONTROLLER:65509
mininet>
```

7. Latency ping (t – 2)

```
mininet> h1 ping h3
PING 10.0.0.3 (10.0.0.3) 56(84) bytes of data.
64 bytes from 10.0.0.3: icmp_seq=1 ttl=64 time=7.40 ms
64 bytes from 10.0.0.3: icmp_seq=2 ttl=64 time=0.222 ms
64 bytes from 10.0.0.3: icmp_seq=3 ttl=64 time=0.144 ms
64 bytes from 10.0.0.3: icmp_seq=4 ttl=64 time=0.149 ms
64 bytes from 10.0.0.3: icmp_seq=5 ttl=64 time=0.141 ms
64 bytes from 10.0.0.3: icmp_seq=6 ttl=64 time=0.145 ms
64 bytes from 10.0.0.3: icmp_seq=7 ttl=64 time=0.359 ms
64 bytes from 10.0.0.3: icmp_seq=8 ttl=64 time=0.139 ms
64 bytes from 10.0.0.3: icmp_seq=9 ttl=64 time=0.143 ms
64 bytes from 10.0.0.3: icmp_seq=10 ttl=64 time=0.175 ms
64 bytes from 10.0.0.3: icmp_seq=11 ttl=64 time=0.137 ms
64 bytes from 10.0.0.3: icmp_seq=12 ttl=64 time=0.151 ms
64 bytes from 10.0.0.3: icmp_seq=13 ttl=64 time=0.161 ms
64 bytes from 10.0.0.3: icmp_seq=14 ttl=64 time=0.166 ms
64 bytes from 10.0.0.3: icmp_seq=15 ttl=64 time=0.280 ms
64 bytes from 10.0.0.3: icmp_seq=16 ttl=64 time=0.139 ms
64 bytes from 10.0.0.3: icmp_seq=17 ttl=64 time=0.166 ms
64 bytes from 10.0.0.3: icmp_seq=18 ttl=64 time=0.164 ms
64 bytes from 10.0.0.3: icmp_seq=19 ttl=64 time=0.168 ms
^C
--- 10.0.0.3 ping statistics ---
19 packets transmitted, 19 received, 0% packet loss, time 18404ms
rtt min/avg/max/mdev = 0.137/0.555/7.398/1.613 ms
mininet>
```

8. Internet Bandwidth (t – 2)

```
naija@VelvetThunder:~$ sudo mnexec -a $(pgrep -f "mininet:h2") iperf3 -c 10.0.0.2
[sudo] password for naija:
Connecting to host 10.0.0.2, port 5201
[  7] local 10.0.0.2 port 43888 connected to 10.0.0.2 port 5201
[ ID] Interval           Transfer     Bitrate         Retr  Cwnd
[  7]   0.00-1.00   sec  4.56 GBytes  39.1 Gbits/sec    0   3.68 MBytes
[  7]   1.00-2.00   sec  5.22 GBytes  44.8 Gbits/sec    0   4.68 MBytes
[  7]   2.00-3.00   sec  5.20 GBytes  44.6 Gbits/sec    0   4.93 MBytes
[  7]   3.00-4.00   sec  5.36 GBytes  46.0 Gbits/sec    1   4.93 MBytes
[  7]   4.00-5.00   sec  4.84 GBytes  41.6 Gbits/sec    1   4.93 MBytes
[  7]   5.00-6.00   sec  5.29 GBytes  45.5 Gbits/sec    3   4.93 MBytes
[  7]   6.00-7.00   sec  5.09 GBytes  43.7 Gbits/sec    0   4.93 MBytes
[  7]   7.00-8.00   sec  5.10 GBytes  43.8 Gbits/sec    1   4.93 MBytes
[  7]   8.00-9.00   sec  4.38 GBytes  37.7 Gbits/sec    0   4.93 MBytes
[  7]   9.00-10.00  sec  4.16 GBytes  35.7 Gbits/sec    0   4.93 MBytes
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Retr
[  7]   0.00-10.00  sec  49.2 GBytes  42.3 Gbits/sec    6             sender
[  7]   0.00-10.00  sec  49.2 GBytes  42.3 Gbits/sec                  receiver

iperf Done.
```

```
mininet> h2 iperf3 -s
-----------------------------------------------------------
Server listening on 5201 (test #1)
-----------------------------------------------------------
Accepted connection from 10.0.0.2, port 43884
[  5] local 10.0.0.2 port 5201 connected to 10.0.0.2 port 43888
[ ID] Interval           Transfer     Bitrate
[  5]   0.00-1.00   sec  4.56 GBytes  39.1 Gbits/sec
[  5]   1.00-2.00   sec  5.22 GBytes  44.8 Gbits/sec
[  5]   2.00-3.00   sec  5.20 GBytes  44.7 Gbits/sec
[  5]   3.00-4.00   sec  5.36 GBytes  46.1 Gbits/sec
[  5]   4.00-5.00   sec  4.85 GBytes  41.7 Gbits/sec
[  5]   5.00-6.00   sec  5.30 GBytes  45.5 Gbits/sec
[  5]   6.00-7.00   sec  5.09 GBytes  43.8 Gbits/sec
[  5]   7.00-8.00   sec  5.11 GBytes  43.9 Gbits/sec
[  5]   8.00-9.00   sec  4.38 GBytes  37.7 Gbits/sec
[  5]   9.00-10.00  sec  4.17 GBytes  35.8 Gbits/sec
[  5]  10.00-10.00  sec   896 KBytes  12.0 Gbits/sec
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate
[  5]   0.00-10.00  sec  49.2 GBytes  42.3 Gbits/sec                  receiver
-----------------------------------------------------------
Server listening on 5201 (test #2)
-----------------------------------------------------------
```